

Conceptual Design for a BSL-3 Suitable Stage-Top Bio-containment chamber



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February 4, 2026

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Chapter 1

Introduction

1.1 Background and Summary

Long term Live-Cell imaging is a very useful research tool when determining long term behavior of cell cultures in various situations. Commercial solutions are unable to provide adequate containment for the use within a Bio-Safety Level 3 (BSL-3) and often require the use of a separate containment enclosure, which introduces additional costs and complexity.

This project focuses on the conceptual design of a low-cost Stage-Top Incubator that is capable of being utilized within a BSL-3 laboratory, without the requirement for additional containment. The primary focus with this project is specific suitability with *Mycobacterium tuberculosis*, however does consider other microbe requirements

1.2 Scope and Limitations

This project is limited to the following:

- Preliminary Research
- Temperature Control
- Compatibility with Multi-Well microtiter plates
- Primary compatibility with *Mycobacterium tuberculosis*.

This project is limited to compatibility with a BSL-3 Lab and does not consider higher containment requirements. The environmental control aspects of the incubator are limited to just temperature control for the preliminary conceptual design. The initial design is limited to compatibility with a multi-well microtiter plate. The use case of this project is limited to that of *Mycobacterium tuberculosis*.

Chapter 2

Research

The conceptual design began with assessing existing commercial products to identify the key environmental control specifications required for the incubator to function correctly. Additionally, this was useful in identifying why the commercial products are incompatible with a BSL-3 lab when used without a separate external biological containment mechanism.

2.1 Zeis/Pecon

The Zeis/Pecon stage top incubator utilities four modules to control the environmental conditions within the incubator stage.

The Pecon Temperature Module S[1] regulates the temperature of the incubator. It has four 8-pin connectors that are used for temperature control of heated components such as the Incubator lid, incubator base, microscope objective and humidifier device. The temperature module is the base module for the entire incubation system, providing electrical power and communications to all connected devices. It directly connects to the CO₂ module to provide power, communication and makes a connection to the gas circulation system to heat the gas flow provided to the incubation chamber. The temperature module requires an external device such as a PC or compatible microscope for operation as it does not have any control panel. The module maintains culture temperature by monitoring and controlling the temperature of the incubator base, lid and optionally the microscope objective. The module infers the culture temperature using pre-set temperatures, or by performing a pre-experiment calibration whereby an identical set-up is placed in the chamber with the included culture temperature sensor to tune the required chamber temperatures.

The Pecon CO₂ Module S[2] attaches to the temperature module for electrical power and communication, and provides CO₂ concentration control to the airflow passed into the incubation chamber. The CO₂ module circulates gas throughout all the connected units and mixes the gas before pumping the resultant mixture into the incubation chamber. The module contains a CO₂ sensor which monitors the concentration of CO₂ in the gas mixture, and controls the release of pure CO₂ into the mixing chamber in order to maintain the desired CO₂ concentration. Due to the method of operation, it requires a >95% purity CO₂ supply.

The Pecon O₂ Module S[3] attaches to the CO₂ module for electrical power and communication. This module functions similarly to the CO₂ module, however it regulates the concentration of O₂ present in the gas mixture by controlling the release of a pure N₂ supply into the air mixture to displace the O₂ present. This method of O₂ displacements means that the O₂ module is only capable of providing hypoxia conditions by lowering O₂ concentration. Due to the method of operation, it requires a >99% purity N₂ supply.

Both the CO₂ module and O₂ module are supplied with a humidifying bottle which functions by bubbling the the gas airflow thorough distilled water to raise the relative humidity (RH). The Pecon Heating Device Humidity S1[4] is an optional module which connects to the Temperature module to heat the humidifying bottle to increase the RH and maintain adequate gas temperature provided to the incubation chamber

The incubator stage includes a heated glass topped lid[5] and a heated insert which accepts mounts for the different culture vessels used in microscopy. The heated insert directly heats the culture medium through physical contact, and the heated lid provides radiative heating and condensation prevention by heating the glass panel directly. The insert supports perfusion experiments utilising two ports which allow tubes to be run to the culture.

2.2 Ibidi

The Ibidi stage top incubator functions similarly to the Zeis/Pecon module, however it makes use of three modules to maintain environmental conditions.

The Heating System Silver Line[6] module regulates the temperature of the incubator. It makes use of 6 independent channels connected via 4 ports, two of which connect to two channels each. These channels are used to monitor and control the temperature of the components used in the system, such as the heated lid and heated plate. The module maintains culture temperature by monitoring and controlling the temperature of the heated lid, heated base and optionally the microscope objective. The module infers the culture temperature using pre-set temperatures, or by performing a calibration whereby an identical set-up is placed in the chamber with the included thermocouple to tune the required chamber settings to meet the desired culture temperature. The temperature module can be operated using the on device buttons and screen, or it can be connected to a PC for control using the free software. The temperature module does not heat the gas mixture.

The Gas Incubation System[7] comes in two different models, the CO₂ - Silver Line and the CO₂/O₂ - Silver Line. The difference between these two systems is the former is only capable of regulating CO₂ concentration, where as the latter has the ability to regulate O₂ concentration as well as CO₂ concentration. The module has three gas inputs for pressurised air, CO₂ and N₂. The CO₂ and N₂ are fed into and mixed with the pressurised air to obtain a mixture with the desired CO₂ and O₂ concentrations. The module then heats the gas mixture before pumping it to the incubation chamber. If the CO₂ only module is used, the N₂ air input is still present on the device, however it is sealed internally and so the module is unable to regulate O₂ concentration. Due to the method of using N₂ to displace O₂ in the mixture, this system is only capable of producing hypoxia environments. The gas incubator module has two output ports which are used in conjunction with the Humidifying column to mix the dry gas with humidified gas to obtain the desired relative humidity as shown in [Figure 2.1](#). The gas incubation module has two electrical connections, one to control the heating of the humidifying column and gas line, and the second to connect to the humidity sensor present in the incubation chamber. The module can be operated using the on device buttons and screen, or it can be connected to a PC for control using the free software.

The Humidifying column[7] works by bubbling the airflow through heated, distilled water to increase

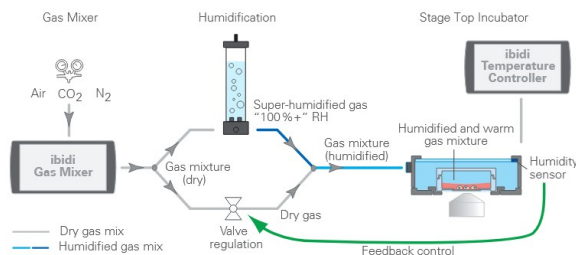


Figure 2.1: Diagram Showing the Humidity Control Mechanism Used by the Ibidi Gas Incubation System [7]

the relative humidity to approximately 100% RH. This is then mixed with dry gas to obtain the desired % RH.

The Ibidi incubation chamber comes in two versions, one version supports culture mediums with Petri dishes and slides, the second version supports multi-well plates. The Petri dish/Slide version of the chamber supports perfusion experiments using two ports which allow tubes to be run to the culture. Both chambers directly heat the culture medium through physical contact and provide radiative heating and condensation prevention by heating the glass lid.

2.3 Tokai-Hit

The Tokai-Hit system functions similarly to the Pecon and Ibidi systems however it utilises a single module to regulate the environmental conditions for temperature and CO₂. Humidity is controlled by a heated water bath inside the chamber, and O₂ control is achieved by utilising an optional module which then overrides all gas control.

The Tokai-Hit STXG/STXF[8] temperature and CO₂ controller regulates both temperature and CO₂ concentration in the incubation chamber. The controller utilises a single connector to control all component temperatures such as the Lid heater, bath heater, stage heater and optional microscope objective heater. The module infers the culture temperature using pre-set temperatures, however it can operate in feedback mode whereby a sensor directly monitors the temperature of the culture medium and automatically adjusts the component temperatures to meet the desired set point. The module can be operated using the built in touch screen or it can be connected to a PC for control using software.

The STXG and STXF models both provide airflow to the incubation chamber, however the STXG model is capable of controlling the CO₂ concentration by feeding supplied CO₂ into the gas mixture until the desired concentration is reached, then pumps the mixture to the incubation chamber. The STXF model is unable to regulate CO₂ concentration, and so requires a pre-mixed pressurised CO₂/air.

The incubator chamber uses a single design that is compatible with most major culture vessels including Petri dishes, slides and multi-well plates. This chamber supports perfusion experiments, drug administration and direct culture temperature measurement via access ports designed into the lid. The humidity inside the incubation chamber is maintained via a water bath that is present inside the chamber, and which is heated by the STXG/STXF module.

The STXG/STXF modules can optionally be connected to a separate CO₂/O₂ unit which overrides

the gas control mixture and allows for the optional control of O₂ concentration.

2.4 Oko-Lab

The Oko-Lab utilises three modules to regulate environmental conditions within the incubation chamber

The Oko-Lab H301-T-UNIT-BL-PLUS[9] unit is responsible for maintaining the thermal conditions within the chamber enclosure. It operates by monitoring and controlling the temperature of the chamber base, lid and optionally the microscope objective. This in turn heats the sample to the desired temperature. This heating unit is capable of sample temperature feedback utilising a thin thermocouple that can be placed in distilled water near the sample to obtain more accurate sample temperature control. Otherwise the unit is capable of using pre-sets/calibrations to infer sample temperature based on enclosure temperatures. The heating unit can be operated via a computer using available software or using optional touch screen accessory. The heating unit is capable of interfacing with the Oko-Lab Bold Line Gas Controllers. The temperature module supports a programmed temperature cycle mode where it cycles between set temperatures.

The CO₂-O₂-UNIT-BL[10] modules come in a variety of formats, differing in the O₂ control ranges they support. The CO₂-O₂-UNIT-BL [0-20; 1-95] variant supports O₂ concentration range from 1% to 95%. This model requires a pressurised supply of CO₂, N₂ and O₂ to obtain the desired gas mixture, alternatively compressed air or an air pump can be connected to the N₂ supply which limits the functionality to just CO₂ concentration control. The CO₂-O₂ BL line all have the capability to directly interface with the Temperature unit to allow communication between the two units.

The HM-ACTIVE[11] operates by passing the gas mixture over heated water to increase the RH. The water temperature can be altered to control the RH of the out flowing gas. The humidity control unit is controlled by the gas mixture unit.

The H301-K-FRAME[12] uses a design that is compatible with most major culture vessels, each using a different sample holder. The chamber is also designed to work with various lids depending on the required application. The chamber supports perfusion experiments and microfluidic experiments. The chamber directly heats the culture medium through physical contact and provides radiative heating and condensation prevention by heating the glass lid

2.5 Shortfalls

2.5.1 Containment

The commercial solutions provided by Zeis/Pecon, ibidi, Tokai Hit and Oko-Lab are all unable to operate independently, without an external containment chamber, within a BSL-3 laboratory due to a fundamental design choice present in all solutions. The Incubation chambers are designed to have an imperfect containment seal to allow the incoming gas flows to passively exhaust out the base and any gaps in the enclosure. This can be seen in the diagram shown in [Figure 2.2](#) which is provided by ibidi.

This means that the exhausting air is unfiltered and may be contaminated with the cells present during the experiment, which can lead to contamination of the surrounding workspace. For this reason, when

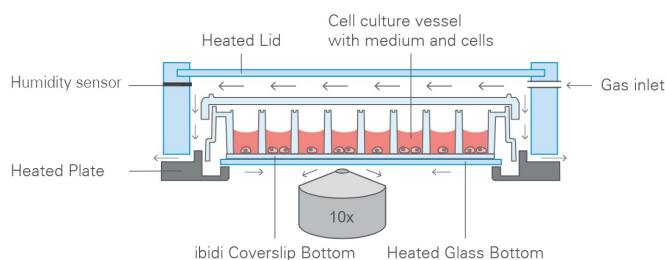


Figure 2.2: Diagram Showing the Gas Flow within the ibidi Multi-Well Incubation Chamber

working with potentially lethal pathogens, it is required that the incubator be placed within a larger containment vessel to ensure that contamination does not occur.

2.5.2 Disinfection and sterilisation

All equipment inside of a BSL-3 lab requires either sterilisation or disinfection depending on the equipment use or properties. Sterilisation is used to kill almost all micro cellular activity on an item and is rated on the Sterility Assurance Level (SAL) which indicates the probability of a single microbe surviving. Disinfection is used to kill enough micro cellular life to achieve a safe enough level required by the laboratory. Examples of sterilisation and disinfection techniques are as follows:

- Sodium hypochlorite(bleach), is useful as a broad spectrum disinfectant and is effective against bacterial spores[13], [14], [15].
- hydrogen peroxide, which is a very effective sterilising chemical[13] and is often used for sterilisation of tool, biological safety cabinets and laboratories[16].
- Quaternary Ammonium Compounds, which provides cost effective but low level disinfection[13], [14], [15].
- Alcohols, which offer an intermediary disinfection [13], however are ineffective against spores[14]
- Autoclave, which provides extremely effective sterilization using a pressurized containment vessel filled with steam heated to approximately 120°C

The aforementioned commercial solutions specify the use of concentrated alcohol solutions such as ethanol or isopropanol, which are not suitable for adequate sterilisation of microbes which produce spores, such as *Mycobacterium tuberculosis*, which is the primary cell considered in this feasibility study. Furthermore, Most of these solutions do not specify whether they are Autoclave compatible, leaving compatibility with autoclave sterilisation uncertain. However, the Tokai hit and Zeis/Pecon incubator specifically mention that they are not compatible with autoclaves.

2.5.3 Key Takeaways

The discussed commercial solutions are incompatible with experiments that require stage top incubation for stand alone, long term, live cell imaging within a BSL-3 lab. This is because of the lack of adequate containment capabilities as well as the incompatibility with disinfection and sterilisation methods required by the microbes of interest.

Chapter 3

Feasibility Analysis

3.1 Commercial Assessment

The commercial solutions from Zeis/Pecon, Ibidi, Tokai-Hit and Oko-lab are assessed for their primary method of operation as well as the environmental control specifications provided by each product.

3.1.1 Zeis/Pecon

The Incubation system provided by Zeis/Pecon utilises four modules for the control of the four major environmental conditions. A temperature unit for thermal control, a CO₂ unit for CO₂ concentration control, an O₂ unit for O₂ concentration control and a humidifying bottle for humidity control.

The heating unit monitors and controls the temperature of the culture sample through heating of the incubation chamber, gas airflow and humidifying bottle.

The CO₂ unit monitors and controls the concentration of CO₂ being pumped into the incubation chamber by controlling the release of pure CO₂ gas into the internal mixing chamber.

The O₂ unit monitors and controls the concentration of O₂ being pumped into the incubation chamber by controlling the release of Pure N₂ gas into the circulating gas in order to displace the O₂ present. Thus can only achieve hypoxia conditions.

The humidifying unit operates by bubbling the gas mixture through heated water to achieve a close to 100% RH to prevent the drying of the culture media.

The Zeis/Pecon incubation system as a whole has the specifications shown in [Table 3.1](#).

Temperature Control Range	Ambient - 60°C (Recommended max 45°C)
Control Accuracy	$\pm 0.15\%$
Control Resolution	Internal Loop: 0.01°C User Parameter: 0.1°C
CO2 Control Range	0.0 - 8.0 vol% (8.1-10 vol% with Uncepecified Reduced Accuracy)
Control Accuracy	± 2 vol%
Control Resolution	Internal Loop: 0.01 vol% User Parameter: 0.1 vol%
O2 Control Range	0.0 - 21.0 vol%
Control Accuracy	± 0.05 vol%
Control Resolution	Internal Loop: 0.01 vol% User Parameter: 0.1 vol%
Operation	PC or Suitable Microscope required
Compatibility	Petri-Dishes, Chambers, Chamber Plates, Chamber Slides, Multi-Well Plates

Table 3.1: Specifications for Zeis/Pecon Incubation System

3.1.2 ibidi

The Incubation system provided by Ibidi utilises three modules to maintain the environmental conditions within in incubation chamber. A temperature unit for thermal control, a CO₂/O₂ unit for CO₂ and O₂ concentration control, and a humidifying column for humidity control.

The heating unit monitors and controls the temperature of the culture sample through heating of the incubation chamber.

The CO₂/O₂ unit monitors and controls the concentration of both CO₂ and O₂ being pumped into the incubation chamber by controlling the release of pure CO₂ and N₂ into the gas mixture. The release of N₂ displaces oxygen present in the air and so the system is only capable of producing hypoxia conditions. The CO₂/O₂ unit is also responsible for the heating of the gas mixture as well as the heating of the humidifying column and gas tube.

The humidification control system works by controlling the mix of dry gas with humidified gas which is obtained through bubbling the dry gas flow through heated distilled water.

The Ibidi incubation system as a whole has the specifications shown in Table 3.2.

Temperature Control Range	Ambient + 5°C - 50°C
Control Accuracy	$\pm 0.2^{\circ}\text{C}$ with $\pm 0.5^{\circ}\text{C}$ uniformity
Control Resolution	User Parameter: 0.1°C
CO₂ Control Range	0.0 - 15.0 vol%
Control Accuracy	± 0.2 vol% + 3% of Reading
Control Resolution	User Parameter: 0.1 vol%
O₂ Control Range	0.5 - 21.0 vol%
Control Accuracy	± 0.2 vol% (typical), ± 0.7 vol% (max)
Control Resolution	User Parameter: 0.1 vol%
Humidity Control Range	Supply RH - 99% RH
Sensor Accuracy	$\pm 2.5\%$
Control Resolution	0.1%
Operation	On Device Controls or Connected PC
Compatibility	Petri-Dishes, Chamber Slides, Multi-Well Plates

Table 3.2: Specifications for Ibidi Incubation System

3.1.3 Tokai Hit

3.1.4 Oko Lab

3.2 Requirements

Describe requirements needed for this project

3.3 Specifications

Describe the specifications that are decided on for this project.

3.4 Feasibility

discuss whether undertaking this project is feasible.

Chapter 4

Conceptual Design

4.1 Design Iterations

These were the iterations and steps taken when conceptually designing the product

4.2 Final Conceptual Design

The final design is the following, and uses the following features.

Chapter 5

Discussion and Results

Chapter 6

Conclusion

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6.1 Recommendations

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